# SysHealth — live demo runbook

Five EC2 sizes, one load, one verdict in dollars. Every command runs from `~/syshealth`.

## 1 The day before

- **The address, once:** `https://syshealth.aryangorde.com` on every `up`, on any network and the audience's phones. $3.65 a month. It prints two records to add at Hostinger (an A record `syshealth` and a CNAME) and waits for the certificate; Ctrl-C is safe, run it again to carry on:

```
./aws.sh domain syshealth.aryangorde.com --profile hackathon
```

- **Rehearse on the network you will present from.** The address is HTTPS, which any network lets through; without it, port 5000, which many campus networks block. Keep a phone hotspot as the way out.
- **Have the fallbacks ready:** the threshold screenshots, the 22 October step screenshots (`~/syshealth-aws-run-2026-10-22/`), the real four-state picture, from the button (`~/syshealth-aws-three-states-button.png`, step 5b), the README's results table, and `npm run dev` working on the laptop (section 5).
- **Changed any code? Push it before rehearsing.** The fleet installs what is on GitHub. Do not run `./aws.sh deploy` on the day: keep the version you rehearsed.

## 2 Forty-five minutes before

The fleet needs about five minutes to boot and calibrate, and you want one rehearsal run in hand before anyone is watching.

### Bring the fleet up

Double-click **SysHealth demo on** on the desktop; it makes the two QR slides too. Or by hand, then the slides:

```
./aws.sh up --profile hackathon
~/syshealth-audience-qr.sh && ~/syshealth-audience-qr.sh --start
```

It names the account, region and cost: about $0.37 an hour, $0.44 with disks, addresses and load balancer (14 of the account's 16 vCPUs). The command asks first; the icon does not.

**Expect:** `All five sizes reporting`, then `https://syshealth.aryangorde.com` opens in the browser, signed in.

### The codes and the slides

The page shows nothing without a code; both are new on every `up`, so make the slides after it, never before.

```
./aws.sh token          # --raw: the token alone, for a script
```

**Expect:** `Server token`: yours, already in the browser; it alone starts and stops tests. `Watch-only code`: for the room; it shows everything and starts nothing.

Slides in `~/syshealth-audience/`: `slide.html` (watch-only) and `slide-start.html` (the token, step 7). Scan both now.

```
./aws.sh status
```

**Expect:** five sizes, `ONLINE yes`, all HEALTHY.

### Rehearse once, from the dashboard

**Load test**, open **Plan it from your users**, keep its defaults (200 users at once normally, 650 at the busiest, app 300 MB, 2 MB each): the form fills with a ramp from 700 MB to 3.1 GB, 6 steps of 60 s, judged against 1600 MB. **Start on 5 instances**. Every **?** explains its part; **Guide**, top right, the page.

Seven minutes: it proves the chain and leaves a result in case the live run fails. Note its verdict (step 5). Step 5b planned? Run it too. Then wait two minutes for the lines to fall. Asked for a code? `./aws.sh open`.

## 3 The demo, about twelve minutes

### Step 0 — Let the room follow along (optional)

Open `~/syshealth-audience/slide.html` full screen (F11): the QR, the address and the watch-only code.

> **Say:** Scan this, or open syshealth.aryangorde.com and type the code. You see what I see, live. Only I can start a test.

### Step 1 — The problem (one minute, no clicking)

> **Say:** Teams pick an instance size by guessing, or by an alarm at 80 percent of memory in use. One way they overpay, the other way the app slows down and nobody knows why. SysHealth measures it: the same load on five sizes at the same moment, then the cheapest size that holds your app, in dollars a month.

### Step 2 — The idle fleet

Five cards, all HEALTHY, a flat chart. Each card's bar is memory in use, marked at 80%. Open **What the number means**. Asked about any part? Its **?** explains it.

> **Say:** This number comes from the Linux kernel itself: the share of time programs waited for memory. Not how full memory is. Under 10 percent is healthy, 10 to 40 is degraded, 40 or more is critical, and so is the whole machine stalling 10 percent of the time. Those lines come from systems that act on this signal in production: Android's low-memory killer, systemd-oomd and Meta's oomd. A state only changes after it has held for 15 seconds, so one spike is not a failure.

### Step 3 — Plan the test from users, and start it

**Load test**, open **Plan it from your users**, keep the defaults. The sums under it show the working; the form below fills itself. **Start on 5 instances**.

> **Say:** Nobody knows their app in megabytes; they know how many people use it at once. 650 at the busiest, 2 megabytes each, on top of 300 for the app: 1.6 gigabytes. The test starts at a normal day, 700 megabytes, and climbs to twice the busiest, on five machines at the same moment. Watch which ones feel it.

### Step 4 — Talk through the six minutes

| Minute | Load | Point at |
|---|---|---|
| 0–1 | 700 MB | t3.micro is 98% in use: nearly full, barely waiting. |
| 1–2 | 1.2 GB | The kernel **kills the test on the micro**; as its reading falls, its card says RAN OUT OF MEMORY. Set **Show** to **Memory in use**: **t3.small reaches 82% in use, past the alarm line, and stays HEALTHY.** |
| 2–3 | 1.7 GB | **Show** back to **Waiting for memory**. t3.small is 99% in use and starts to wait. Click it for **Where the memory is**. |
| 3–4 | 2.1 GB | The kernel **kills the test on the small**. t3.medium is 69% in use, 0% waiting. |
| 4–6 | 2.6–3.1 GB | **t3.medium fills up and stays HEALTHY** (95% in use at 3.1 GB). t3.large and t3.xlarge are flat. |

Minutes 0–4 are the 1 November run, 4–6 the 19 October run at the same loads. States judge a one-minute average, so a card turns up to a minute after its step begins. Use the quiet stretches for the Prometheus question (section 4).

> **Say:** At 1.2 gigabytes, look at the small. 82 percent of its memory is in use, past the line where the usual alarm rings. But nothing is waiting, so nothing is slowed down. That is the difference between memory in use and memory pressure.

### Step 5 — The verdict

Open **Load test**. **Which size to choose** leads the results. Quote the rehearsal's numbers if they differ from these.

> **Say:** Choose t3.medium, 32 dollars 70 a month. The small would save 16 dollars 35, but it was full at 1.7 gigabytes and killed at 2.1, and we asked for 25 percent spare over 1.6. The large and the xlarge cost two and four times as much for room this app does not use. A sizing decision made from measurement, and said in money. Its last line says what it did not test: the same 4 gigabytes on Graviton or AMD, t4g.medium and t3a.medium, for about half. Those go in the next run before anyone switches.

Then point at the **Pressure at each load level** chart: where a size's line crosses the degraded line is the most it can hold. Change **Your app at its busiest** or **Keep spare** and the answer is worked out again, without another test.

### Step 5b — All three states at once (optional, six minutes)

**Load test**, **Show all three states**, **Start**. From about 1½ minutes to the end: micro CRITICAL, small DEGRADED, medium, large and xlarge HEALTHY. Not red by step 3? Show the screenshot. (2.1 GB twice: rounding.)

> **Say:** A real server has other work that keeps running, so the small is slowed but still working: degraded. The medium and large are at 100 percent CPU, and nothing waits.

### Step 6 — Beyond the dashboard (optional, one minute)

The same verdict from the terminal, instantly, from the test that just ran:

```
URL=http://<dashboard-ip>:5000      # ./aws.sh status prints it
export SYSHEALTH_TOKEN=$(./aws.sh token --raw)
python3 cli.py --server $URL report
```

> **Say:** The same test runs from the command line, so it can gate a build: it fails when the size the app needs has changed. Prometheus can scrape everything at /metrics, and an AI agent can run the experiment through an MCP server.

### Step 7 — Let the room run one (optional, after the verdict)

Open `slide-start.html`. Its QR holds the server token: whoever scans it can start a test, or stop one. One runs at a time; the token dies at `./aws.sh down`.

> **Say:** Scan this one and run your own: Load test, Plan it from your users, your app's numbers, Start.

## 4 Questions you will get

### “Why is the micro's number so low at the end?”

> **Say:** It is not coping, it failed. The load did not fit in its memory and swap, so the kernel killed it. Pressure fell because the work stopped. That is why the card says ran out of memory, and the results mark every step after it as critical.

Proof, live, if they want it: the kernel's own record, with times that match the chart.

```
./aws.sh ssh t3.micro 'sudo journalctl -k --since "20 min ago" | grep -i killed'
```

### “Where does 2 MB per user come from?”

> **Say:** It is the one number you measure: your app's memory with no users, and with a known number at once; the difference over the users. The plan turns users into megabytes; the test measures everything after that.

### “What if my app needs more than the biggest size?”

> **Say:** The fleet goes up to t3.xlarge, 16 gigabytes. If none fits, the verdict says so rather than guess, and the same command tests bigger sizes with --types.

### “Why did nothing show DEGRADED?”

> **Say:** The test load is the only work on the box, so when it waits, everything waits, and the whole-machine rule fires as soon as degraded would. Give it other work and it shows, as step 5b does.

No time: `~/syshealth-aws-three-states-button.png`, 23 October.

### “Does it work on a real app?”

> **Say:** Yes. We ran a language model, Gemma 4 in llama.cpp, on all four sizes. The micro could not load it, the small was killed twelve times, and the medium answered all 39 requests at 80 percent in use with nothing waiting.

### “How is this different from Prometheus and Grafana?”

> **Say:** Prometheus watches. It can graph this same kernel counter, but it never applies a load, never knows what load caused a reading, and never says what to buy. SysHealth runs the same load on every size at once, works out how much each holds before the app slows down, and names the cheapest that fits. They work together too: everything, the verdict included, is served at /metrics for Prometheus to scrape.

### “Why not just watch memory in use?”

> **Say:** It does not say whether it hurts. At 95 percent in use the medium was stalled 0.03 percent of the time; at 97 the micro was stalled 13. Users feel the waiting.

### “A peak test said t3.small. Why does the ramp say t3.medium?”

> **Say:** A peak test only asks whether each size can hold the app at 1.6 gigabytes, not how much more it could take. The ramp measures the room left, and with 25 percent spare asked for, the small does not have it.

### “Could this run in production?”

> **Say:** The measurement side, yes: Prometheus can scrape it today, seeing it needs a code, and tests need the token. History is kept in memory, so a real deployment would need a database first.

### “What is it built on?”

> **Say:** Python and Flask. The agent reads the kernel's pressure files every five seconds; Terraform builds or destroys the fleet in one command.

# 5 If something breaks

### The page asks for a code

The browser has none, or one from an earlier fleet. `./aws.sh open` opens it signed in; `./aws.sh token` prints both codes again.

### The address does not open

Just after `up`, the load balancer takes a minute or two; `up` checks and says so. Type the `https://`: plain http does not answer. Meanwhile the second address `./aws.sh status` prints works from the laptop.

### The dashboard will not load

Without the address: almost always your address changed, a new network or a mobile provider moving you overnight. Or the network blocks port 5000. It looks exactly like a dead server.

```
./aws.sh up --profile hackathon      # moves the firewall to this address
```

**Expect:** `0 added, 3 changed, 0 destroyed`. Still nothing: use the phone hotspot and run it again.

A phone on the laptop's Wi-Fi can open it too; on mobile data it is blocked.

### The page stops updating in the middle of a test

Seen once, 1 November, cause unknown. Reload once. Still nothing: open the address on a phone on mobile data, which tells your network from the server. The test runs on the machines either way; the rehearsal's result is your fallback. Then look:

```
./aws.sh ssh dashboard 'uptime; free -m; sudo journalctl -u syshealth-server -n 30'
```

### A size is missing

```
./aws.sh status                      # which sizes are not reporting
./aws.sh ssh t3.small 'sudo tail -30 /var/log/syshealth-bootstrap.log'
./aws.sh ssh t3.small 'sudo journalctl -u syshealth -n 30'
```

### Everything stays HEALTHY under load

The load fits. Check the **Right now** tile moves at all, then raise **To** in the form.

### Nothing on AWS works and you must present anyway

A local simulation needs no AWS and no network: the real dashboard, a simulated fleet.

```
sudo apt install -y python3-flask python3-requests   # first time only
npm run dev                                          # http://127.0.0.1:5000
```

Say it is simulated (every page says **Demo data**), then show the README's real results.

# 6 Afterwards

Save the evidence first (history lives in the dashboard's memory; URL as in step 6), then destroy the fleet:

```
export SYSHEALTH_TOKEN=$(./aws.sh token --raw)
python3 cli.py --server $URL report --save rightsize-$(date +%F).json
curl -s -H "Authorization: Bearer $SYSHEALTH_TOKEN" "$URL/series?limit=720" \
  > demo-evidence-$(date +%F).json
./aws.sh down && ./aws.sh status      # or double-click SysHealth demo off
```

**Expect:** `Nothing is up`: billing has stopped. Until then the fleet bills about $0.44 an hour, about $10.50 a day.